

Turbomachinery CFD Validation Portfolio

Case study of the 19 planned cases: evidence, findings and decisions needed

Lead CFD Engineer
for review by the Principal CFD Lead

5 October 2026

Purpose. Present every planned turbomachinery case that is not yet validated: what the rig is, what we have built, what the solver showed, and what it takes to validate it against measurements.

Decisions requested from the Principal CFD Lead

1. Approve the validation order for the next five cases (slide: Recommended plan).
2. Approve the compute needed for the transient Timisoara case (several vortex-rope periods, 1–4 days on 10 cores).
3. Approve data-access requests: Radiver (signed RWTH licence), MEXICO (IEA Wind), Pelton (NTNU), SHF impeller (Lille), Turbine-99 (Vattenfall).
4. Agree that axialTurbine stays a verification case only (no experiment exists).

Every rig fact was checked against the primary source on 5 Oct 2026 (reference list at the end). Images are our own renders unless the caption names a source.

Where we are

- ▶ 19 planned cases; none validated yet.
- ▶ 6 have run: 2 ran to the end (NREL, axialTurbine), 4 crashed or did not converge (Francis-99, LS89, Rotor 37, Timisoara).
- ▶ 4 are set up only (Turbine-99, Aachen, Pelton, Mini-ORC); 9 are planned (PPTC cavitation, SHF, 7 backlog).
- ▶ Measured data is on disk for Francis-99, LS89, Rotor 37 and PPTC.

Recommendation: fix and validate Francis-99, PPTC cavitation, NREL, LS89 and Rotor 37, in that order.

Key technical findings

- ▶ LS89 and Rotor 37 ran **laminar**: the SST settings were present but inactive.
- ▶ LS89 fields before the crash are non-physical (blade p up to $4.6 \times P_0$).
- ▶ The Francis-99 mesh is the **part-load** point; the case ω (35.12 rad/s) differs from the measured 34.87 rad/s.
- ▶ NREL: torque at 5 m/s has the wrong sign; 1 checkMesh failure.
- ▶ SHF is a separate rig, not a duplicate of ERCOFTAC.

- ▶ 19 cases: every planned case not yet compared with experiment. The 4 validated cases (ERCOFTAC pump, FDA blood pump, PPTC open water, single-channel pump) are not repeated.
- ▶ One slide per case, same rows on every slide: geometry, domain and mesh, fluid, physics and solver, rotation model, boundary conditions, operating point, validation data, references, status.
- ▶ Rig facts (geometry, operating point, data) were checked on 5 Oct 2026 against the primary sources listed on each slide and in the reference list at the end. Set-up values for built cases come from the case dictionaries and logs. Set-ups not yet built are marked *planned*: they are proposals, not facts.

Status		Validation data	
Ran, not validated	solver ran, no comparison yet solver crashed or did not converge dictionaries only, no mesh or run in the plan, nothing built	data in hand	on this machine
Ran, did not finish		public, to download	open, to download
Setup only		not available	not public or none
Planned, not started			

Overview of the 19 cases

#	Case	Machine	Physics	Rotation model	Validation data
1	Francis-99	Francis turbine	incompr. RANS	MRF + mixingPlane	data in hand public, to download data in hand data in hand public, to download (paper)
2	NREL Phase VI	HAWT wind turbine	incompr. RANS	MRF	
3	VKI LS89	transonic turbine vane	compressible	none (cascade)	
4	NASA Rotor 37	transonic compressor	compressible	MRF	
5	Timisoara swirl generator	swirl gen. / runner	incompr. URANS	sliding mesh (AMI)	
6	axialTurbine (6 variants)	axial hydro stage	incompr. RANS/URANS	SRF, MRF, mixingPlane, AMI	not available (no experiment)
7	Turbine-99	Kaplan draft tube	incompr. RANS	none (rotating cone wall)	not available (raw data offline)
8	Aachen 1.5-stage	axial gas turbine	compressible	MRF + mixingPlane	not available (SU2 result only)
9	Pelton (NTNU)	impulse turbine	VOF two-phase	sliding mesh (ACMI)	efficiency only, CAD on request
10	Mini-ORC (TU Delft)	radial-inflow turbine	real-gas compressible	MRF + mixingPlane	not available (no measurements)
11	PPTC cavitating	propeller	cavitation (VOF)	MRF or sliding mesh	data in hand published (PIV, wall pressure) public, to download open, licence needed published, geometry incomplete public, to download
12	SHF impeller	centrifugal pump (air model)	incompr. RANS	MRF or frozen rotor	
13	NASA Stage 35	axial compressor stage	compressible	MRF + mixingPlane	
14	Radiver	centrifugal compressor	compressible	MRF + mixingPlane	
15	Krain impeller	centrifugal compressor	compressible	MRF	
16	NASA HECC	centrifugal compressor	compressible	MRF + mixingPlane	data public, CAD not consortium access per chosen rig
17	NASA SDT R4	turbofan fan	compressible	MRF + mixingPlane	
18	MEXICO rotor	HAWT wind turbine	incompr. RANS	MRF	
19	Wells turbine / OWC	wave-energy air turbine	oscill. flow, VOF chamber	sliding mesh or MRF	

1. Francis-99 (NTNU model Francis turbine, 2nd workshop)

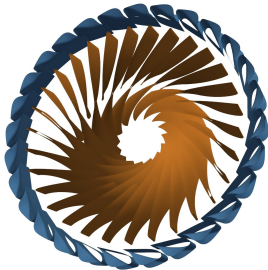
Status: **Ran, did not finish**: 1000 iterations, residuals $\sim 10^{-2}$; serial only

Validation data: **data in hand**

Geometry	NTNU Waterpower Lab model: 1:5.1 model of the Tokke prototype: spiral casing, 14 stay vanes, 28 guide vanes, runner 15 blades + 15 splitters, draft tube. Mesh used: guide-vane angle 6.72°
Domain and mesh	Single passage of guide vane and runner (periodic) + full draft tube; workshop mesh 1.89M cells, 141 patches; non-orth. 72.6° , skewness 3.94
Fluid	Water, incompressible, Newtonian
Physics and solver	Steady RANS, <code>simpleFoam</code> , standard $k-\varepsilon$; PBiCGStab for p , relaxation 0.1, upwind U
Rotation model	MRF runner, case value $\omega = -35.123$ rad/s (335.4 rpm; measured at PL: 34.87 rad/s); mixingPlane at GV–runner and runner–draft tube; 8 periodic <code>ggi</code> pairs (TurboWG libraries)
Boundary conditions	Inlet (GV inflow): prescribed velocity, radial -1.428 , tangential -2.14 m/s, axial 0; outlet: static pressure; walls: no-slip with wall functions
Operating point	Part load : workshop data give $\alpha = 6.72^\circ$ for PL (BEP 9.84° , HL 12.46°); PL $H \approx 11.87$ m, $Q \approx 0.14$ m ³ /s
Validation data	NTNU 2nd-workshop data, downloaded: steady pressure at VL2 (vaneless space), DT5, DT6 (draft tube); steady velocity; PIV on 3 draft-tube lines (<code>f99w2-exp-*</code>)
References	Trivedi, Cervantes, Gandhi, Dahlhaug (2013) <i>J. Fluids Eng.</i> 135(11):111102; Francis-99 2nd workshop data (NTNU, Dec 2016)
Gap / next step	Continue to convergence (ε and <code>mixingPlane</code> flux still oscillate), set ω to the PL value, then compare pressure and PIV

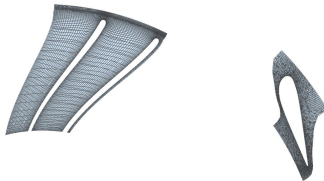
1. Francis-99: geometry, mesh and solver state

Geometry



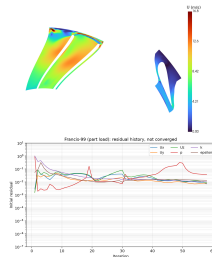
Runner (15+15 blades) and 28 guide vanes;
single passage copied around the axis for
display

Mesh



Workshop hex mesh, mid-span cut through
runner and guide-vane passages (1.89M cells)

Solution at iteration 1000 (not
converged)



$|U|$ at mid-span; residuals stall near 10^{-2}

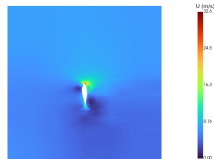
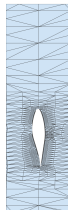
Assessment. Geometry, mesh and measured data (pressure, LDV, PIV, part load) are all in hand: this is the cheapest case to validate. **Risk:** ε and mixingPlane flux still oscillate; the case runs serial only (parallel crashes in *ggi*). **Action:** set ω to the measured 34.87 rad/s, continue to a steady state, then compare VL2/DT5/DT6 pressure and the three PIV lines.

2. NREL Phase VI (Unsteady Aerodynamics Experiment, wind turbine)

Status: **Ran, not validated**: 7 wind speeds, 2000 iterations each

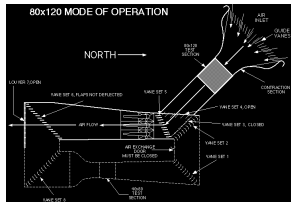
Validation data: **public, to download**

Geometry	2-bladed stall-regulated HAWT, S809 airfoil, $D = 10.058$ m, tip pitch 3° , NASA Ames 24.4×36.6 m tunnel; lofted in Python from the chord/twist tables of TP-500-29955 (root transition, rounded tip cap)
Domain and mesh	Full rotor in a box; STL $\rightarrow$ snappyHexMesh, 1.31M cells; 1 checkMesh test failed (146 skewed faces, max 10.8)
Fluid	Air, incompressible, $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$
Physics and solver	Steady RANS, simpleFoam, $k-\omega$ SST, TI 1% (assumed)
Rotation model	MRF rotor zone, $\omega = 7.54 \text{ rad/s}$ (72 rpm)
Boundary conditions	Inlet: fixed velocity 5–25 m/s; outlet: $p = 0$, inletOutlet U; far field: slip; blades: no-slip
Operating point	Sequence S (upwind, rigid, 3° tip pitch, 5–25 m/s), 0° yaw subset; 72 rpm; $U_\infty = 5, 7, 10, 13, 15, 20, 25 \text{ m/s}$
Validation data	Shaft torque, root flap moment; C_p from 22 taps at each of 30, 46.6, 63.3, 80, 95% span, integrated to C_n , C_t ; only TP-500-29955 is here, measured data (TP-500-29494, DOE Wind Data Hub) to download
References	Hand et al. (2001) NREL/TP-500-29955; Simms et al. (2001) NREL/TP-500-29494
Gap / next step	5 m/s torque has the wrong sign; fix mesh quality, then compare torque and C_p



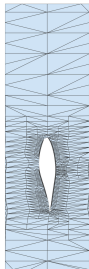
2. NREL Phase VI: geometry, mesh and results

Test set-up (source)



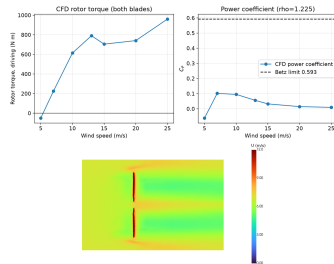
NASA Ames 80×120 ft tunnel layout.
Source: Hand et al.,
NREL/TP-500-29955

Mesh



snappyHexMesh, blade section at 80%
span (1.31M cells)

Results (7 wind speeds)



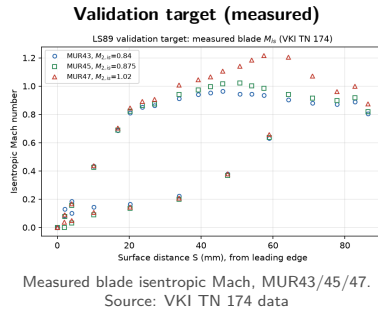
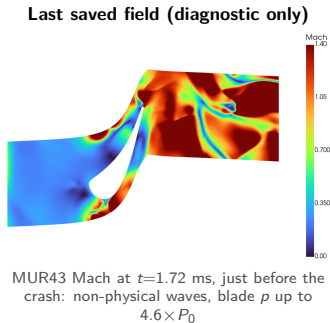
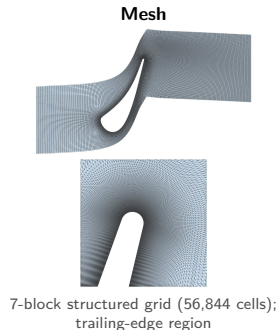
Assessment. All 7 wind speeds ran to 2000 iterations. **Risk:** the 5 m/s torque has the wrong sign, and one checkMesh test fails (146 skewed faces); the measured torque and C_p data (TP-500-29494) still has to be downloaded. **Action:** fix mesh quality, rerun 5 and 7 m/s, compare torque and C_p at the five stations.

3. VKI LS89 (highly loaded transonic turbine guide-vane cascade)

Status: **Ran, did not finish**: 7 conditions, all crashed at 63–93% of target time Validation data: **data in hand**

Geometry	Linear cascade of the VKI LS89 vane: chord 67.647 mm, pitch 57.5 mm, stagger 55° ; 2D section
Domain and mesh	One blade passage, periodic; 7-block Plot3D grid converted (Python 3 rewrite of p3d2gmsh), 56,844 hex cells, 16 stitched interfaces, front/back empty
Fluid	Air, perfect gas, <code>hePsiThermo</code> , Sutherland viscosity
Physics and solver	Compressible, transient, density-based <code>rhoCentralFoam</code> ; run laminar (SST block present but inactive). <code>rhoSimpleFoam</code> failed on 13 μm trailing-edge cells
Rotation model	None (stationary cascade); <code>cyclic</code> pitchwise periodicity
Boundary conditions	Inlet: total pressure 1.435 MPa (MUR43) and total temperature 420 K; outlet: static pressure 0.904 MPa; blade: no-slip, adiabatic
Operating point	MUR43–MUR49: P_0 1.433–1.608 MPa, $T_0 = 420$ K, $M_{2,is} = 0.84, 0.875, 1.02$, $Re_2 = 10^6$
Validation data	Blade-surface isentropic Mach number and pressure (digitised <code>mach.lay</code> , <code>pressure.lay</code>); heat-transfer data in TN 174
References	Arts, Lambert de Rouvroit, Rutherford (1990) VKI Technical Note 174; AboutFlow benchmark test case 2 (QMUL)
Gap / next step	Restart from checkpoints, fix the floating-point crash in the thermo library, add SST, compare surface Mach

3. VKI LS89: mesh, crash diagnostics and validation target



Assessment. The solution was laminar (SST inactive) and degraded before the floating-point crash, so the snapshot cannot be compared with experiment. **Action:** switch on SST, restart from a clean initial field with a lower Courant number, and run to a periodic state; then compare surface M_{is} for MUR43 and MUR47.

4. NASA Rotor 37 (isolated transonic axial compressor rotor)

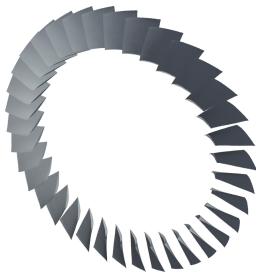
Status: **Ran, did not finish**: crashes after $\sim 10^{-6}$ s

Validation data: **data in hand**

Geometry	36 MCA blades, tip speed 454 m/s, rig tip clearance 0.356 mm (0.45% span; case uses 0.35 mm); blade rebuilt from the MERIDL geometry file (own parser)
Domain and mesh	Single passage, 10° periodic; 214–258k cells; 42 skewed faces (max 7.3)
Fluid	Air, perfect gas
Physics and solver	Compressible, rhoPimpleFoam; run laminar (SST block present but inactive), ramped ω
Rotation model	MRF , 17,188.7 rpm (1800 rad/s); periodic sides as cyclicAMI pair
Boundary conditions	Inlet: total pressure 101,325 Pa, total temperature 288.15 K; outlet: static pressure 213,391 Pa (estimated from design PR); hub, shroud, blade: no-slip
Operating point	Design: PR 2.106, adiabatic efficiency 0.889, 20.19 kg/s; speedline from choke to near stall
Validation data	Speedline (PR, efficiency vs mass flow), radial total pressure/temperature and flow angle at stations 1 and 4, laser anemometry; Excel archive Rotor37_exp.zip downloaded
References	Reid & Moore (1978) NASA TP-1337; Suder & Celestina (1996) <i>J. Turbomach.</i> 118(2):218–229; Denton (1997) <i>J. Therm. Sci.</i> 6:1–13; Dunham (1998) AGARD-AR-355
Gap / next step	Improve tip and periodic mesh, start at low back-pressure and ramp, turn on SST, then run a speedline

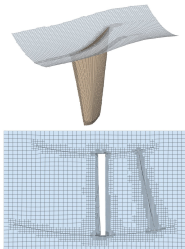
4. NASA Rotor 37: geometry, mesh and validation target

Geometry



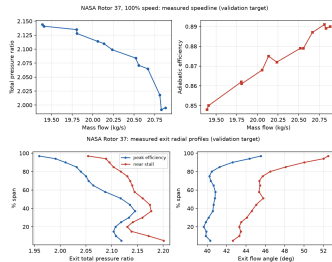
36 blades rebuilt from the MERIDL file (single blade copied for display)

Mesh



Blade and hub surface mesh; axial cut
(214–258k cells)

Validation target (measured)



100% speed: speedline and exit radial profiles. Source: NASA data archive (Suder)

Assessment. The solver crashes after about 10^{-6} s; it also ran laminar, and the outlet pressure was an estimate. Measured speedline and radial profiles are in hand. **Action:** rebuild the tip-gap mesh (42 skewed faces), use SST, start at low back-pressure and ramp up, then march along the speedline.

5. Timisoara swirl generator (Politehnica Univ. Timisoara)

Status: **Ran, did not finish**: stable to $t = 0.238$ s, then SIGFPE Validation data: **public, to download** (paper)

Geometry	4 leaned struts → 13 guide vanes → free 10-blade runner → convergent–divergent test section (reproduces the vortex rope of a Francis draft tube at part load)
Domain and mesh	Full 360°; shipped mesh 2.70M hex cells, 18 patches; rotating zone 708,050 cells; max non-orth. 75° at AMI faces
Fluid	Water, incompressible, $\nu = 10^{-6}$ m ² /s
Physics and solver	Transient URANS, pimpleFoam, RNG $k-\varepsilon$
Rotation model	Sliding mesh : dynamicMotionSolverFvMesh (solid-body rotation), 3 cyclicAMI interfaces; $\omega = 96.342$ rad/s (920 rpm). Ported from foam-extend GGI
Boundary conditions	Inlet: axial velocity 1.69765 m/s; outlet: fixedMean $p = 0$; runner: movingWallVelocity; other walls: no-slip
Operating point	30 L/s; free runner measured at 920 rpm (design 870 rpm)
Validation data	LDV mean velocity profiles in the cone (used by Petit et al. 2011) and unsteady wall pressure (Bosioc et al. 2012); not on this machine
References	Susan-Resiga et al. (2006) <i>J. Fluids Eng.</i> 128:177–189; Bosioc et al. (2012) <i>J. Fluids Eng.</i> 134(8):081104; Petit et al. (2011) <i>IJFMS</i> 4(1):199–208
Gap / next step	Restart from checkpoint, run several rope periods (1–4 days on 10 cores), average and compare LDV



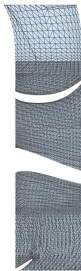
5. Timisoara swirl generator: geometry, mesh and transient results

Geometry



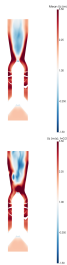
Struts, 13 guide vanes, 10-blade free runner, test cone

Mesh



AMI interface region (2.70M hex cells)

Results to $t = 0.238$ s



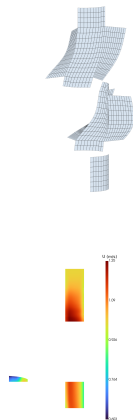
Mean and instantaneous $|U|$ (sliding mesh, 3 cyclic AMI interfaces)

Assessment. The sliding-mesh method works (AMI weights 0.97–1.0; $3.1\times$ speed-up on 10 cores), but the long run hit SIGFPE at 0.238 s. LDV comparison needs several vortex-rope periods: 1–4 days on 10 cores. **Decision needed:** approve that compute, after a restart from the last checkpoint with a smaller time step.

6. axialTurbine (TurboWG teaching case, 6 rotor–stator methods)

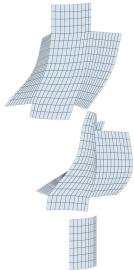
Status: **Ran, not validated**: all 6 methods agree within 1.7% at the runner outlet Validation data: **not available**

Geometry	Synthetic axial hydro stage: guide vanes → 5-blade runner → draft tube
Domain and mesh	Single passage (cases 1–5) or full 5-blade annulus (case 6); 1,400 / 3,700 / 18,500 cells
Fluid	Incompressible, $\nu = 10^{-5} \text{ m}^2/\text{s}$ (non-dimensional teaching set-up)
Physics and solver	Steady (SRFSimpleFoam, simpleFoam) and transient (SRFPimpleFoam, pimpleFoam); standard $k-\varepsilon$
Rotation model	All of them : SRF steady and transient; MRF + cyclicPeriodicAMI; MRF + mixingPlane; rotating mesh with AMI, 1 blade and 5 blades; $\omega = -10 \text{ rad/s}$
Boundary conditions	Inlet: axial velocity 1 m/s; outlet: zero-gradient U, fixed p ; walls: no-slip
Operating point	Single design point
Validation data	No experiment exists for this geometry; method-vs-method comparison only (axial U within 1.7%, flow angle within 4.3°)
References	Nilsson, H., rotating-machinery training, OpenFOAM Workshops OFW7 (2012) and OFW10 (2015); Beaudoin et al. (2014) mixingPlane, IOP Conf. Ser.
Gap / next step	Cannot be validated; keep as a verification case for the rotation models



6. axialTurbine: geometry, mesh and cross-method verification

Geometry and mesh



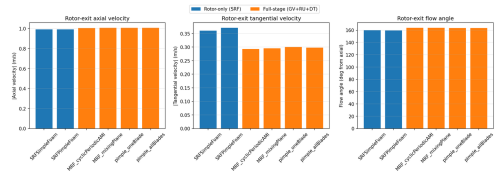
Guide vanes, 5-blade runner, draft tube

Mesh cut



Blade-to-blade mesh

Six rotor-stator methods compared



Axial velocity at the runner outlet: 6 methods agree within 1.7%

Assessment. All six rotation models (SRF, MRF + periodic AMI, MRF + mixingPlane, rotating mesh with 1 and 5 blades) give the same answer within 1.7%. No measurements exist for this teaching geometry. **Decision needed:** keep it as a verification case for rotation models only.

7. Turbine-99 (Hölleforsen Kaplan draft tube, Vattenfall)

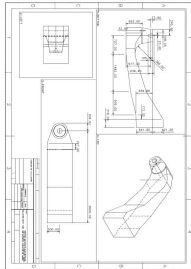
Status: **Setup only**: dictionaries, no mesh

Validation data: **not available** (raw data offline)

Geometry	1:11 model of the Hölleforsen sharp-heel elbow draft tube, runner $D = 0.5$ m: runner cone, elbow, rectangular diffuser. CAD file not downloadable (host dead since 2010)
Domain and mesh	<i>Planned</i> : draft tube only, from the inlet section below the runner to the outlet; hex mesh to be built from the KBwiki geometry description
Fluid	Water, incompressible
Physics and solver	<i>Planned</i> : steady RANS, simpleFoam, $k-\omega$ SST (skeleton present)
Rotation model	None for the flow; rotating runner-cone wall (595 rpm)
Boundary conditions	<i>Planned</i> : inlet = measured LDV axial and tangential mean/RMS profiles at section Ia (radial component missing, to be modelled); outlet: static pressure; cone: rotating wall; walls: no-slip
Operating point	Case T (on-cam, 60% load): $H = 4.5$ m, 595 rpm, $Q = 0.522\text{--}0.528$ m ³ /s, $Re_D \approx 1.75M$
Validation data	Wall pressure recovery and LDV velocity at sections Ia, Ib, II, III; only described in ERCOFTAC KBwiki AC6-07, raw files not public
References	Andersson (2000) Turbine 99 workshop proceedings, ISSN 1402-1536; ERCOFTAC KBwiki AC6-07 (Karlsson); Nilsson & Page (2005), Turbine-99 III, Porjus
Gap / next step	Obtain CAD and inlet/validation data from Vattenfall or Luleå; otherwise not feasible

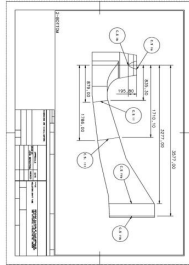
7. Turbine-99: rig geometry and measurement sections (source figures)

Geometry



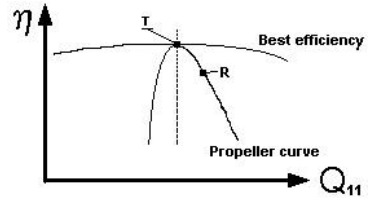
Hölleforsen draft-tube geometry definition.
Source: ERCOFTAC KBwiki AC6-07

Measurement sections



LDV and pressure section locations (Ia, Ib, II, III). Source: KBwiki AC6-07

Operating points



Propeller curve with test cases T and R. Source: KBwiki AC6-07

Assessment. The case is well documented, but the CAD file and raw LDV/pressure files are not downloadable (host offline since about 2010). No mesh can be built from the figures alone. **Decision needed:** request the data from Vattenfall / Luleå, or drop the case.

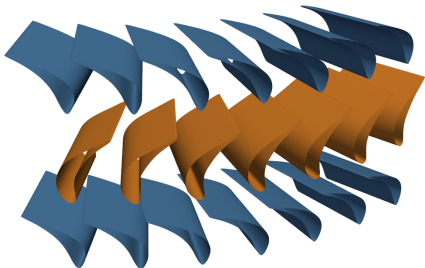
8. Aachen 1.5-stage axial turbine (RWTH Aachen, IST)

Status: **Setup only**: 3 meshes converted from SU2, not run Validation data: **not available** (SU2 result only)

Geometry	Stator 1 → rotor → stator 2; modified VKI stator vanes, Traupel rotor profile; untwisted, constant hub/tip; for CFD the stator count is raised from 36 to 41 so every row has 41
Domain and mesh	Single passage per row, 8.78° ($360^\circ/41$) periodic; 3 OpenFOAM meshes converted from the SU2 coarse mesh
Fluid	Air, ideal gas ($\gamma = 1.4$, $R = 287.058$), Sutherland viscosity
Physics and solver	<i>Planned</i> : steady compressible RANS, rhoSimpleFoam, SA or $k-\omega$ SST (SU2 used SA + wall functions)
Rotation model	MRF rotor, $\omega = -366.52$ rad/s (3500 rpm); mixingPlane between the 3 rows (compressible + mixingPlane untested in v2412)
Boundary conditions	Inlet: total pressure 158,245 Pa, total temperature 308.26 K, axial; outlet: static pressure 110,051 Pa; TI 2.5%, $\nu_t/\nu = 100$; walls: no-slip
Operating point	Total-to-static PR ≈ 1.44 , motion Mach 0.35
Validation data	RWTH measurements not public; converged SU2 mixing-plane solution available for code-to-code comparison
References	Gallus, Zeschky & Hah (1995) <i>J. Turbomach.</i> 117(4):562–570; SU2 Aachen turbine tutorial; Oliani et al. (2022), new OpenFOAM turbomachinery solver applied to a 1.5-stage turbine
Gap / next step	Test compressible + mixingPlane; request IST data or accept code-to-code comparison only

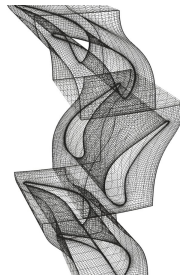
8. Aachen 1.5-stage turbine: geometry and converted meshes

Geometry



Stator 1 – rotor – stator 2, 41 blades per row (7 passages shown)

Mesh



Three single-passage meshes converted from the SU2 tutorial (about 174k cells in total)

Assessment. Meshes are ready, but RWTH measurements are not public; only a code-to-code comparison with the SU2 mixing-plane solution is possible. The compressible solver with `mixingPlane` is untested in v2412. **Decision needed:** accept code-to-code only, or request IST data.

9. Pelton turbine (NTNU Waterpower Lab reference design)

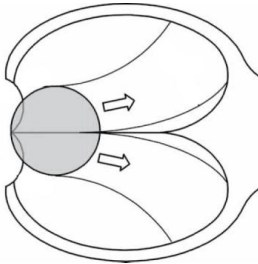
Status: **Setup only:** dictionaries, no mesh

Validation data: efficiency published; CAD on request

Geometry	Horizontal single-jet Pelton: runner $D = 513$ mm (pitch radius 256.7 mm), 23 buckets, bucket width 114.2 mm, jet diameter 35 mm; geometry and data "available upon request"
Domain and mesh	<i>Planned:</i> nozzle + jet region (static) and runner sector or full runner (rotating), casing
Fluid	Water and air, two-phase, immiscible, isothermal
Physics and solver	<i>Planned:</i> transient VOF, <code>interFoam</code> , $k-\omega$ SST
Rotation model	Sliding mesh with <code>cyclicACMI</code> (jet crosses the interface); <code>overset overInterDyMFoam</code> as fallback. MRF not suitable
Boundary conditions	<i>Planned:</i> nozzle inlet $\alpha = 1$, jet velocity $\sqrt{2gH} \approx 37$ m/s; casing outlets atmospheric total pressure; buckets: moving walls; $\omega \approx 65$ rad/s (placeholder, not sourced)
Operating point	$H = 70$ m, best-efficiency speed to be confirmed
Validation data	Measured efficiency at $H = 70$ m (max 77.75%) and visual inspection (Solemslie & Dahlhaug 2014); NTUA casing data as a cross-check
References	Solemslie & Dahlhaug (2012) IOP EES 15:032005, (2014) IOP EES 22:012004; Petley & Aggidis (2019) <i>IJFMS</i> 12(4); Rygg (2013) NTNU MSc thesis, <code>interDyMFoam</code> + AMI
Gap / next step	Request bucket geometry from NTNU; highest compute cost of all cases (several revolutions at VOF Courant limit)

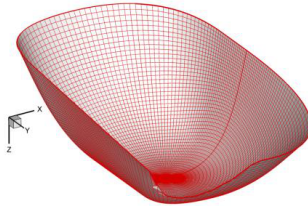
9. Pelton (NTNU reference design): rig geometry (source figures)

Jet-bucket concept



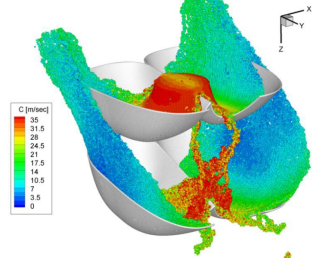
Bucket and jet. Source: Solemslie & Dahlhaug (2012)

Bucket geometry



Parametric bucket surface. Source: Solemslie & Dahlhaug (2012)

Reference CFD



Published free-surface simulation of jet and buckets. Source: Solemslie & Dahlhaug (2012)

Assessment. Main parameters are published ($D = 513$ mm, 23 buckets, jet 35 mm, $H = 70$ m), but the bucket geometry is “available upon request”. This is the most expensive case: VOF, sliding mesh with ACMI, several revolutions. **Decision needed:** request the geometry from NTNU before any build.

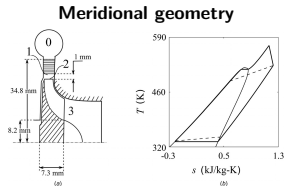
10. Mini-ORC radial-inflow turbine (TU Delft, ORCHID)

Status: **Setup only**: draft dictionaries only

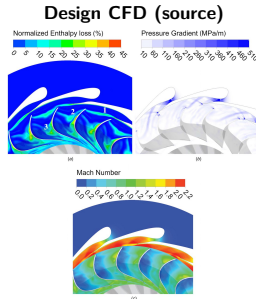
Validation data: **not available** (no measurements yet)

Geometry	Single-stage radial-inflow turbine, 12 nozzle vanes, rotor $D_2 = 51.5$ mm, 12 kW; meanline and CFD design published, no 3D blade coordinates
Domain and mesh	<i>Planned</i> : one nozzle passage + one rotor passage, periodic
Fluid	Siloxane MM (hexamethyldisiloxane), non-ideal (dense) gas
Physics and solver	<i>Planned</i> : compressible RANS with a real-gas equation of state (Peng–Robinson or CoolProp tables); no validated NICFD solver in v2412, needs a new solver family
Rotation model	MRF rotor + mixingPlane (or frozen rotor) at the nozzle–rotor interface; 98,000 rpm
Boundary conditions	<i>Planned</i> : inlet total pressure 1810 kPa, total temperature 573 K; outlet static pressure 44.3 kPa
Operating point	Volume ratio 48, nozzle exit angle 78° , design η_{ts} 83% (meanline); ORCHID rig limit 25 bar, 300°C
Validation data	No published turbine measurements; only the design-method CFD of de Servi et al.
References	de Servi, Burigana, Pini, Colonna (2019) <i>J. Eng. Gas Turbines Power</i> 141(9):091021; Head et al. (2016) ORCHID preliminary design
Gap / next step	Not feasible until a real-gas solver is built and measurements are published

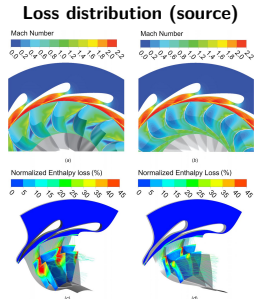
10. Mini-ORC radial-inflow turbine: design geometry (source figures)



Stator and rotor meridional section;
T-s diagram. Source: de Servi et al.
(2019), CC BY



Published Mach number and loss fields. Source:
de Servi et al. (2019), CC BY



Published entropy-loss fields. Source: de Servi et
al. (2019), CC BY

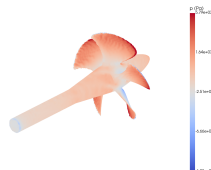
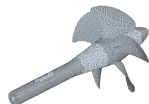
Assessment. Only design CFD is published; there are no turbine measurements yet and no 3D blade coordinates. OpenFOAM v2412 has no validated real-gas (NICFD) solver. **Decision needed:** park until measurements are published or a real-gas solver is qualified.

11. PPTC VP1304 cavitating propeller (SVA Potsdam, SMP'11)

Status: **Planned, not started**: open water validated; cavitation templates only

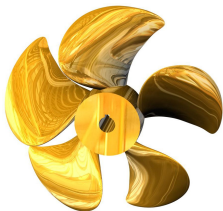
Validation data: **data in hand**

Geometry	5-blade right-handed CPP VP1304, $D = 0.25$ m, $P_{0.7}/D = 1.635$, $A_E/A_0 = 0.779$; cavitation-tunnel hub cap (differs from open-water rig)
Domain and mesh	<i>Planned</i> : K15A tunnel section 600×600 mm or open domain; reuse the validated M3 snappyHexMesh set-up (1.16M cells) with tip refinement
Fluid	Water + vapour (Schnerr–Sauer); measured $\rho = 997.5$ kg/m ³ , $\nu = 9.4 \times 10^{-7}$ m ² /s, $p_v \approx 2.8$ kPa
Physics and solver	<i>Planned</i> : interPhaseChangeFoam, VOF + Schnerr–Sauer, $k-\omega$ SST
Rotation model	MRF for the first pass; sliding mesh (cyclicAMI) for transient cavity shapes
Boundary conditions	<i>Planned</i> : inlet velocity from J ; outlet pressure set from σ_n ; propeller: rotating no-slip; tunnel walls: slip
Operating point	SVA test cases 2.3.1/2.3.2/2.3.3: $J = 1.019 / 1.269 / 1.408$, $\sigma_n = 2.024 / 1.424 / 2.000$, $n = 25$ s ⁻¹
Validation data	Cavity sketches, photos and videos; cavitating K_T , K_Q ; LDV planes (report 3754); all in references/
References	SVA reports 3752, 3753, 3754 (Potsdam, 2011); Gaggero & Villa (2018) <i>J. Mar. Sci. Appl.</i> 17(1):1–20
Gap / next step	Strongest new case: geometry, mesh and data are ready



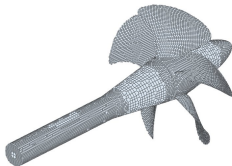
11. PPTC cavitating propeller: geometry, mesh, open-water result and cavitation target

Geometry (source)



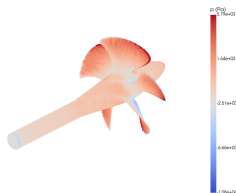
VP1304. Source: SVA report 3753

Mesh



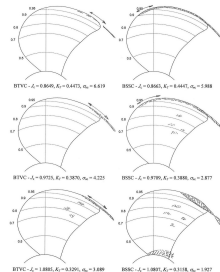
Our open-water blade surface mesh

Open-water result



Blade pressure, $J = 0.802$
(validated: $K_T - 5.4\%$)

Cavitation target (measured)



Observed cavity extents. Source: SVA report 3753

Assessment. The strongest new case: geometry, a validated open-water set-up, and measured cavity sketches, photos and LDV data are all in hand. **Action:** add Schnerr–Sauer cavitation (interPhaseChangeFoam) to the validated M3 mesh with tip refinement; start with test case 2.3.1 ($J = 1.019$, $\sigma_n = 2.024$).

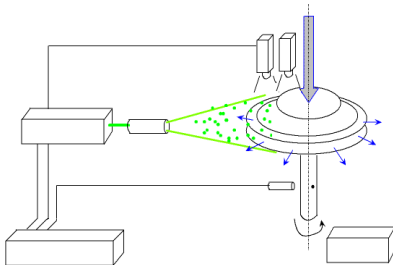
12. SHF centrifugal pump impeller (Société Hydrotechnique de France, air model)

Status: **Planned, not started**: correction, this is *not* the ERCOFTAC U3 rig Validation data: published (PIV, wall pressure)

Geometry	Radial pump impeller, 7 blades, outlet radius 0.2566 m, outlet height 38.5 mm, outlet blade angle 22.5° ; tested with a vaneless diffuser, vaned diffusers (8 channel-type or 11 cascade-type vanes) and an industrial volute
Domain and mesh	<i>Planned</i> : full 360° impeller + diffuser (rotor–stator interaction is not periodic)
Fluid	Air at 25°C (low-speed, treated as incompressible)
Physics and solver	<i>Planned</i> : steady RANS (simpleFoam), then transient sliding mesh for blade-position-resolved comparison
Rotation model	MRF (frozen rotor at fixed blade position) or sliding mesh (cyclicAMI)
Boundary conditions	<i>Planned</i> : inlet flow rate; outlet static pressure; walls: no-slip
Operating point	1710 rpm, design flow $Q_n = 0.337 \text{ m}^3/\text{s}$, tip speed 45.95 m/s; tests at 0.30–1.63 Q_n
Validation data	2D PIV of impeller outlet and vaneless diffuser (Wuibaut et al. 2001); mean and unsteady wall pressure at 114 taps in one blade passage (Godinho Patricio 1999); directional-probe traverses with vaned diffuser and volute (El Hajem et al. 1998)
References	Wuibaut et al. (2001) <i>Proc. IMechE A</i> 215(6); Godinho Patricio (1999) PhD thesis, Univ. Lille; El Hajem et al. (1998) <i>La Houille Blanche</i> 7; Atif, Benmansour, Bois (2010) <i>Int. J. Rotating Mach.</i> 706043
Gap / next step	Earlier “duplicate” decision was wrong; full blade geometry availability to be confirmed with Arts et Métiers / Lille (LML)

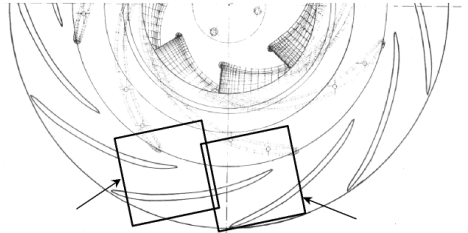
12. SHF impeller: rig and measurement set-up (source figures)

PIV set-up



PIV arrangement on the air model. Source: Atif et al., CFM 2017

Impeller and diffuser



Impeller (7 blades) with vaned diffuser and PIV windows. Source: Atif et al., CFM 2017

Assessment. A genuine, separate test case with published PIV and wall-pressure data (1710 rpm, $Q_n = 0.337 \text{ m}^3/\text{s}$).

Risk: the full blade geometry is not confirmed as open. **Decision needed:** request the geometry from Lille (LML / Arts et Métiers).

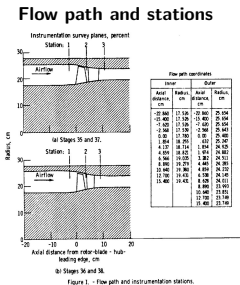
13. NASA Stage 35 (single-stage transonic axial compressor)

Status: **Planned, not started**: backlog

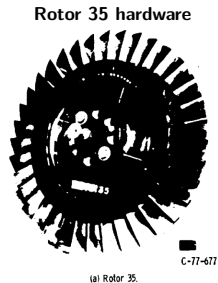
Validation data: **public, to download**

Geometry	Rotor 35 (36 blades, tip solidity 1.3, aspect ratio 1.19) + stator 35 (46 blades, aspect ratio 1.26); one of four inlet stages (35–38) of NASA TP-1337
Domain and mesh	<i>Planned</i> : single passage per row, periodic; structured blade mesh
Fluid	Air, perfect gas
Physics and solver	<i>Planned</i> : steady compressible RANS, rhoSimpleFoam, $k-\omega$ SST
Rotation model	MRF rotor + mixingPlane to the stator
Boundary conditions	<i>Planned</i> : inlet standard total pressure and temperature; outlet static pressure swept along the speedline
Operating point	Design: 17,188.7 rpm, tip speed 455 m/s, 20.19 kg/s, stage PR 1.82, stage adiabatic efficiency 0.828
Validation data	Overall performance maps and radial surveys of total pressure, temperature and flow angle (NASA TP-1337, TP-1338)
References	Reid & Moore (1978) NASA TP-1337; Reid & Moore (1978) NASA TP-1338 (blade-element data)
Gap / next step	Do after Rotor 37: same speed, flow and tip speed, so set-up and blade parser carry over

13. NASA Stage 35: flow path and hardware (source figures)



Flow path and instrumentation stations, stages 35 and 37. Source: NASA TP-1337, Fig. 1



Rotor 35 blade row. Source: NASA TP-1337, Fig. 18(a)

Assessment. Fully public geometry and data. It shares speed, flow and tip speed with Rotor 37, so the blade parser and set-up carry over. **Action:** start once Rotor 37 runs stably.

14. Radiver (centrifugal compressor with vaned diffuser, RWTH Aachen / MTU)

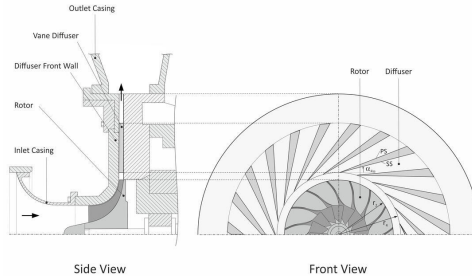
Status: **Planned, not started**: backlog

Validation data: open test case, signed licence needed

Geometry	Semi-open impeller with 15 backward-curved blades (no splitters); wedge vaned diffuser with 23 vanes, adjustable in two degrees of freedom; rig commissioned by MTU, run at RWTH IST
Domain and mesh	<i>Planned</i> : one impeller passage + one diffuser passage (non-matching pitch handled by the interface)
Fluid	Air, perfect gas
Physics and solver	<i>Planned</i> : steady compressible RANS, rhoSimpleFoam, $k-\omega$ SST
Rotation model	MRF impeller + mixingPlane at the impeller–diffuser gap (frozen rotor as a check)
Boundary conditions	<i>Planned</i> : inlet total pressure and temperature; outlet static pressure or mass flow
Operating point	Design speed 35,200 rpm, design pressure ratio 4; several diffuser vane settings
Validation data	Steady and unsteady measurements (performance, PIV, unsteady pressure) for several diffuser configurations; released as an RWTH IST open test case, available after signing a licence agreement
References	Ziegler, Gallus & Niehuis (2003) <i>J. Turbomach.</i> 125(1):173–182 (Part I) and 183–192 (Part II); RWTH IST “Open Testcases” page
Gap / next step	Sign the IST licence agreement and request the package (testcases@ist.rwth-aachen.de)

14. Radiver: rig (source figure)

Centrifugal compressor stage



Radiver stage. Source: RWTH Aachen IST, Open Testcases page

Correction. The data is released as an open test case but needs a **signed licence agreement** (testcases@ist.rwth-aachen.de); it is not a free download. **Decision needed:** approve signing the licence.

15. Krain impeller (high-pressure-ratio centrifugal compressor, DLR)

Status: **Planned, not started**: backlog

Validation data: published; geometry incomplete

Geometry	Unshrouded impeller, 24 full blades, 30° backsweep at exit, vaneless diffuser; published geometry does not fully define the blade (reconstruction needed)
Domain and mesh	<i>Planned</i> : one blade passage with tip clearance + vaneless diffuser, periodic
Fluid	Air, perfect gas
Physics and solver	<i>Planned</i> : steady compressible RANS, rhoSimpleFoam, $k-\omega$ SST
Rotation model	MRF impeller (single rotating passage)
Boundary conditions	<i>Planned</i> : inlet total pressure and temperature; outlet mass flow or static pressure
Operating point	Design 22,363 rpm, 4.0 kg/s, total rotor pressure ratio 4.7
Validation data	Laser-two-focus (L2F) velocity measurements upstream, along the passage and downstream of the impeller; performance data
References	Krain (1988) "Swirling impeller flow", <i>J. Turbomach.</i> 110(1):122–128; Eisenlohr, Krain, Richter, Tiede (2002) ASME GT2002-30394
Gap / next step	Geometry reconstruction risk; lower priority than Radiver and HECC. No open-licence figure located

16. NASA HECC (High-Efficiency Centrifugal Compressor, NASA Glenn CE-18)

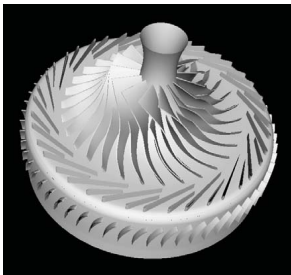
Status: **Planned, not started**: backlog

Validation data: **public, to download** (public archive)

Geometry	Impeller with 15 main/splitter blade pairs (backswEEP 32–42°), exit diameter 17.0 in; vaned diffuser with 20 main/splitter vane pairs; 90° bend; 60 exit guide vanes
Domain and mesh	<i>Planned</i> : one impeller, one diffuser and one EGV passage, periodic, tip clearance 0.012 in
Fluid	Air, perfect gas
Physics and solver	<i>Planned</i> : steady compressible RANS, rhoSimpleFoam, $k-\omega$ SST
Rotation model	MRF impeller + mixingPlane at impeller–diffuser and diffuser–EGV interfaces
Boundary conditions	<i>Planned</i> : inlet total pressure 11 psia (rig), total temperature; outlet static pressure swept to the map
Operating point	Design corrected speed 21,789 rpm, $U_2 = 1615$ ft/s; design-intent PR 4.85 (14.7 psia) / 4.80 (11 psia), measured 4.68; inlet corrected flow 10.85 lbm/s measured; work factor 0.79 design, 0.81 measured
Validation data	Public data archive: stage and subcomponent performance maps, static pressures, 3-hole probe surveys at diffuser exit
References	Braunscheidel, Welch, Skoch, Medic, Sharma (2014) AIAA paper, NTRS 20140016535; Harrison et al. (2024) NASA/TM-20240003692/REV1, NTRS 20250000831
Gap / next step	Best open centrifugal-compressor data set; download the archive first

16. NASA HECC: hardware and facility (source figures)

Impeller



Splittered impeller (15 pairs). Source: Braunscheidel et al. (2014), NASA

Test facility



NASA Glenn CE-18 facility. Source: Braunscheidel et al. (2014), NASA

Assessment. The best open centrifugal-compressor data set (geometry, maps, surveys in a public archive). **Action:** download the archive and check geometry completeness before committing.

17. NASA SDT fan R4 (Source Diagnostic Test)

Status: **Planned, not started**: backlog

Validation data: data public, CAD not

Geometry	1/5-scale high-bypass fan, 22 in diameter, 22 blades, designed by GE Aircraft Engines; OGV sets: Baseline 54 vanes, Low Count 26 vanes, Low Noise; blade CAD not public
Domain and mesh	<i>Planned</i> : one rotor passage + one OGV passage, periodic, with tip clearance
Fluid	Air, perfect gas
Physics and solver	<i>Planned</i> : steady compressible RANS, rhoSimpleFoam, $k-\omega$ SST
Rotation model	MRF rotor + mixingPlane to the OGV (rotor-alone case: MRF only)
Boundary conditions	<i>Planned</i> : inlet total conditions of the 9×15 ft tunnel; bypass outlet static pressure
Operating point	Design corrected speed 12,657 rpm, corrected tip speed 1,215 ft/s, design stage PR 1.47; approach, cutback and takeoff points
Validation data	Rotor-alone and stage performance, LDV flow measurements (NASA Glenn 9×15 ft Low Speed Wind Tunnel)
References	Hughes, Jeracki, Woodward, Miller (2005) NASA/TM-2005-211681 (AIAA-2002-2426); Van Zante et al., SDT LDV report, NTRS 20030066234
Gap / next step	Not feasible without the blade geometry; confirm access with NASA Glenn / GE

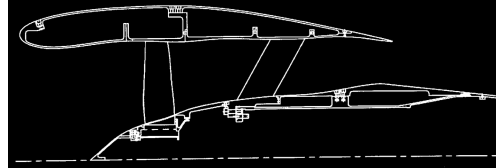
17. NASA SDT fan R4: hardware (source figures)

Fan rig in the 9×15 ft tunnel



R4 fan model. Source: Hughes et al., NASA/TM-2005-211681

Configuration



Model cross-section. Source: Hughes et al., NASA/TM-2005-211681

Assessment. Performance and LDV data are public, but the blade geometry is a GE design and not published. **Decision needed:** drop unless NASA Glenn / GE release the geometry.

18. MEXICO rotor (EU project MEXICO, IEA Wind Task 29 Mexnext)

Status: **Planned, not started**: backlog

Validation data: consortium access

Geometry	3-bladed rotor, $D = 4.5$ m, DU91-W2-250, RISØ-A1-21 and NACA 64-418 sections, with nacelle; DNW LLF 9.5×9.5 m open test section
Domain and mesh	<i>Planned</i> : full rotor (or 120° periodic) in the open-jet test section
Fluid	Air, incompressible
Physics and solver	<i>Planned</i> : steady RANS, simpleFoam, $k-\omega$ SST
Rotation model	MRF rotor for axial flow (sliding mesh for yawed cases)
Boundary conditions	<i>Planned</i> : inlet uniform velocity; outlet static pressure; open-jet / far-field boundary; blades: no-slip
Operating point	Axial flow, 424.5 rpm, pitch -2.3° , $U_\infty = 10, 15, 24$ m/s (tip speed ratio 10, 6.7, 4.2)
Validation data	148 Kulite pressure sensors at 25, 35, 60, 82, 92% radius; rotor loads; PIV wake data
References	Schepers & Snel (2007) ECN-E-07-042 (MEXICO final report); Schepers, Boorsma, Munduate (2014) <i>J. Phys. Conf. Ser.</i> 555:012089
Gap / next step	Confirm data access with IEA Wind Task 29/47

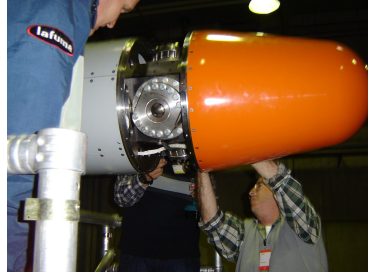
18. MEXICO rotor: hardware (source figures)

Blades



MEXICO blades. Source: Schepers & Snel (2007), ECN-E-07-042

Rotor and nacelle



Rotor installation. Source: Schepers & Snel (2007), ECN-E-07-042

Assessment. Very rich data (148 pressure sensors, PIV), strong OpenFOAM precedent, but access goes through IEA Wind Task 29/47. **Decision needed:** approve a data-access request.

19. Wells turbine / oscillating water column (wave energy)

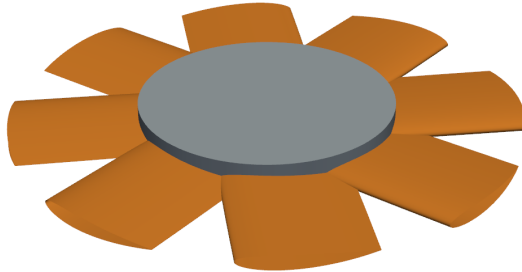
Status: **Planned, not started**: backlog

Validation data: depends on chosen rig

Geometry	Self-rectifying axial air turbine with symmetric (typically NACA 00xx) blades set in the plane of rotation, above an OWC chamber; no single canonical open rig
Domain and mesh	<i>Planned</i> : turbine passage (periodic); optional OWC chamber with wave flume
Fluid	Air (turbine); water + air for the chamber
Physics and solver	<i>Planned</i> : turbine: transient incompressible RANS, <code>pimpleFoam</code> , $k-\omega$ SST, oscillating inflow; chamber: VOF (<code>interFoam</code>)
Rotation model	MRF for quasi-steady characteristic curves; sliding mesh for the oscillating cycle
Boundary conditions	<i>Planned</i> : inlet time-varying (sinusoidal) velocity or pressure; outlet static pressure; blades: no-slip
Operating point	Flow-coefficient sweep at constant speed (rig-specific)
Validation data	Torque and pressure-drop coefficients vs flow coefficient from the chosen rig (see reviews)
References	Raghunathan (1995) <i>Prog. Aerosp. Sci.</i> 31(4):335–386; Shehata et al. (2017) <i>Int. J. Energy Res.</i> 41:6–38; 16 kW Wells/OWC design for Ecuador (2025) <i>Processes</i> 13:2349 (waves in OpenFOAM)
Gap / next step	Choose a rig with open geometry and measurements first

19. Wells turbine: illustrative geometry

Generic Wells rotor (illustrative)



Generated here for illustration only: 8 NACA 0012 blades, chord in the plane of rotation. Not a specific rig.

Assessment. No canonical open rig; a rig with open geometry and measurements must be chosen first. **Decision needed:** keep in backlog.

1. **Check the active turbulence model, not only the dictionary.** LS89 and Rotor 37 carried SST settings but ran with `simulationType laminar`.
2. **Inspect fields before the crash time.** LS89 looked fine in residuals but its fields were already non-physical.
3. **Match the operating point to the data file, not the folder name.** The Francis-99 guide-vane angle in `p1.txt` identified the mesh as part load; the case ω still differs from the measured value.
4. **Mesh quality gates first.** NREL (1 failed check) and Rotor 37 (42 skewed faces) need fixing before any comparison; PPTC passed Mesh OK and validated.
5. **Parallel decomposition is case-dependent.** Francis-99 crashes in parallel (`ggi/mixingPlane`); Timisoara runs $3.1\times$ faster on 10 cores.
6. **Confirm data access before meshing.** Turbine-99, Pelton, SDT, Radiver and MEXICO are limited by data or geometry access, not by CFD.

Order	Case	Work needed	Data
1	Francis-99 (part load)	set measured ω , converge, compare pressure, LDV, PIV	in hand
2	PPTC cavitating	add Schnerr–Sauer to the validated M3 mesh; case 2.3.1 first	in hand
3	NREL Phase VI	fix mesh, rerun 5/7 m/s, compare torque and C_p	download
4	VKI LS89	SST, clean restart, periodic state, compare M_{is}	in hand
5	NASA Rotor 37	tip mesh, SST, back-pressure ramp, speedline	in hand
–	Timisoara	restart; several rope periods (1–4 days, 10 cores)	paper (LDV)
–	Stage 35, HECC	after Rotor 37 runs stably	public
–	Data requests	Radiver, MEXICO, SHF, Pelton, Turbine-99	access

Not recommended now: Mini-ORC (no real-gas solver, no data), SDT R4 (geometry proprietary), axialTurbine (verification only), Krain (geometry rebuild).

Data in hand or open

- ▶ Francis-99 (part load)
- ▶ PPTC cavitating
- ▶ VKI LS89
- ▶ NREL Phase VI
- ▶ NASA Rotor 37
- ▶ Stage 35, HECC; Radiver (licence)

Possible, with cost or access

- ▶ Timisoara (1–4 days compute)
- ▶ SHF impeller (geometry access)
- ▶ Aachen (code-to-code only)
- ▶ MEXICO (consortium data)
- ▶ Krain (geometry rebuild)
- ▶ Wells (choose rig)

Blocked

- ▶ axialTurbine (no experiment)
- ▶ Turbine-99 (CAD, data offline)
- ▶ Pelton (geometry on request)
- ▶ Mini-ORC (no solver, no data)
- ▶ SDT R4 (CAD proprietary)

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